

许仁杰

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教育背景

复旦大学数学科学学院	应用数学 博士	上海	2017.09-2024.01
- 主修方向: 科学计算与矩阵算法; 主修课程: 张量计算、随机算法、信号处理、量子计算、人工智能与大型优化问题求解等; 2024 年上海市优秀毕业生 2019 年获复旦大学第二批卓越博士生促进计划资助			
复旦大学数学科学学院	应用数学 学士	上海	2013.09-2017.06
复旦大学大数据学院	数据科学 学士 (辅修)	上海	2014.09-2017.06
- 主修课程: 高等代数、偏微分方程、概率论与数理统计、纠错码的代数理论、实变函数与泛函分析、数值方法与科学计算基础等; 2016 年丘成桐大学生数学竞赛分析优胜奖 2015 年“深圳杯”数学建模竞赛二等奖 2014 年大学生数学竞赛二等奖			

职业经历

牛津-香港城市大学联合中心-智能多维数据分析研究中心有限公司(香港)	博士后研究员	香港	2024.02-至今
- 高效准确的 Attention 机制加速大模型推理与训练: 「算法创新」: 设计高效准确的稀疏 Attention 机制 (ELFATT、DuSA), 突破视觉 Transformer 中 softmax 注意力计算复杂度为 $O(N^2)$ 的瓶颈, 将其降维至近似线性, 并严格证明近似误差上界。 「性能提升」: ELFATT 在多媒体视觉任务中相较原始 softmax 注意力实现 4-7 倍加速, 同时保持模型性能; 在边缘设备 (如 NVIDIA Jetson) 上于 5W 至 60W 功耗模式下, 依然相较现有 SOTA 模型实现 1.6-2.0 倍加速; DuSA 在训练与推理阶段显著提速, 训练完成后性能优于标准注意力机制 (VSA)。在多个 benchmark 任务中 (如 Long Range Arena、图像分类、语义分割、目标检测、文本生成视频、长文本理解) 均取得领先表现。 「成果落地」: 技术成果已发表于 NeurIPS 2025、ACM MM 2025、NeurIPS 2025 Workshop ER, 并投稿至 ICLR 2026; 已获得中国发明专利 (202510108088.7)、香港短期专利 (P00222.HKS) 与美国非临时专利 (P00222.US)。该研究成果广泛适用于语音处理、视觉生成和扩散模型推理加速, 具备高度拓展潜力。 - 该研究成果广泛适用于语音处理、视觉生成和扩散模型推理加速, 具备高度拓展潜力。 - 具身智能中的无人机动态路径规划系统研发: 「算法设计」: 面向具身智能中的“感知-决策-行动”闭环系统, 提出**演化协作差分进化 (ECDE) **框架, 融合环境变化感知、自适应变异与个体协作策略, 实现多目标路径的实时优化与高效重规划。 「性能验证」: 在高仿真城市 3D 模拟环境中测试, 覆盖高楼遮挡、人口密度分布与动态风险场景。ECDE 相较传统方法如 NSGA-II 等, 在路径效率与能耗方面提升超过 20%, 在避障能力与解集多样性方面表现稳定。 「成果落地」: 相关论文已被 IEEE TVT 接收, 算法已应用于智慧城市无人机模块, 具有良好的可解释性与可扩展性, 适用于多种具身智能场景中的自主决策与导航任务。			
华为技术有限公司(上海) 2012实验室	理论研究部实习生	上海	2020.07-2021.01
- 智能交通信号修复算法研究: 「算法创新」: 针对高缺失率的多源交通数据, 设计了基于 ADMM 与共轭梯度预处理的鲁棒低秩补全模型, 并引入时空核正则项以捕捉动态路网特性, 有效突破传统非凸优化方法在大规模数据下的收敛瓶颈。 「性能提升」: 在广州道路、杭州轨道交通、西雅图高速公路等真实数据集上实现超过 95% 的补全精度, 较基准模型提升约 15%; 收敛速度提升近 10 倍, 支持千万级矩阵的实时修复应用。 「成果落地」: 核心成果发表在 JCAM (JCR Q1) 期刊, 并成功应用于华为通信网络监控模块, 在高峰时段有效提升通行效率 12.6%, 展现出较强的实际部署价值。			

博士研究亮点

博士论文:《两类超复数矩阵的分解算法与应用研究》

- 研究方向: 计算数学、矩阵优化与高效算法设计
- 核心结论:
 - 「算法创新」: 针对超复数矩阵 (对偶数与四元数) 结构下的计算挑战, 提出多种具备封闭解与数值稳定性的高效低秩分解算法 (如 DUTV、DQR、QUTV、QCUR), 突破传统 SVD 在复数扩展空间中的应用限制。
 - 「运算优化」: 本研究构建融合随机采样与矩阵分块策略的高效算法框架, 在保障分解精度的同时显著降低时间与内存复杂度。该框架支持 GPU 并行与边缘设备部署, 具备在深度学习任务中作为前处理与推理加速模块的潜力, 适用于视觉智能与医学信号等大规模数据场景。
 - 「成果落地」: 相关研究成果已发表于 PJO、CAM、AML、NFAO、LMA 等多个国际期刊, 共计五篇论文, 内容涵盖超复数矩阵分解的理论推导与实际应用。算法已成功应用于脑功能成像、彩色视频分析等任务中, 展示出良好的跨领域泛化能力。

个人技能

- 电脑技能:** 熟悉主流编程语言 (如 Python、C、MATLAB) 及深度学习开发流程, 具备一定的模型训练与部署经验。了解 PyTorch 框架和 GPU 并行计算, 具备处理高维数据与优化模型性能的能力, 曾参与模型在嵌入式设备上的应用开发。
- 语言能力:** 普通话、英语。
- 特長與愛好:** 户外山地运动 (中国登山协会户外山地运动指导员)、足球 (中国足球协会三级裁判)、赛艇。

Renjie Xu

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EDUCATION

Fudan University, School of Mathematical Science	Ph.D. in Applied Mathematics	Shanghai	Sep.2017-Jan.2024
Major Areas: Scientific Computing and Matrix Algorithms; Core Courses: Tensor Computation, Stochastic Algorithms, Signal Processing, Quantum Computing, Artificial Intelligence, and Large-scale Optimization, etc. - Honors: Outstanding Graduate of Shanghai (2024) Recipient of the 2019 Fudan University Excellent Doctoral Student Advancement Program			
Fudan University, School of Mathematical Science	B.Sc. in Applied Mathematics	Shanghai	Sep.2013-Jun.2017
Fudan University, School of Big Data	B.Sc. in Data Science(Minor)	Shanghai	Sep.2014-Jun.2017
- Relevant Coursework: Advanced Algebra, Partial Differential Equations, Probability and Mathematical Statistics, Algebraic Coding Theory, Real and Functional Analysis, and Numerical Methods for Scientific Computing - Honors: 2016 Yau College Student Mathematics Competition – Merit Award in Analysis (2016); Shenzhen Cup Mathematical Modeling Competition – Second Prize (2015); National College Mathematics Competition – Second Prize (2014)			

WORK EXPERIENCE

Oxford-CityU Joint Centre for Intelligent Multidimensional Data Analysis Ltd.	Postdoctoral Research Fellow Hong Kong	Feb. 2024 - Present
Efficient and Accurate Attention Mechanism for Large Language Model Acceleration: [Algorithm Innovation] : Developed high-efficiency linear/sparse attention mechanisms (ELFATT, DuSA) that reduce the quadratic complexity $O(N^2)$ of softmax attention in transformers to almost linear complexity while providing rigorous approximation error bounds. [Performance Enhancement] : ELFATT achieves 4–7× speedups over vanilla softmax attention on different vision tasks with minimal performance loss. On edge platforms (e.g., NVIDIA Jetson), demonstrated 1.6–2.0× acceleration across 5W–60W power modes. DuSA offers a significant acceleration in both training and inference phases, even outperforming vanilla attention after training. Achieved SOTA performance across long range arena, image classification, semantic segmentation, object detection, text-to-video generation, and long context understanding. [Implementation and Impact] : Published in NeurIPS 2025, ACM MM 2025, NeurIPS 2025 Workshop (ER), and submitted to ICLR 2026. Technology patented in China (CN: 202510108088.7), Hong Kong (P00222.HKS), and the US (P00222.US). Results applicable to speech processing, visual generation, and diffusion model inference acceleration, demonstrating high extensibility. UAV Path Planning in Embodied Intelligence Systems: [Algorithm Innovation] : Proposed an Evolutionary Collaborative Differential Evolution (ECDE) framework for multi-objective path optimization and real-time replanning in embodied AI systems, integrating environmental adaptation, mutation self-adaptation, and cooperative individual learning strategies to complete the perception–decision–action loop efficiently. [Performance Enhancement] : Tested in a high-fidelity 3D urban simulator including occlusions, population density, and dynamic risk sources. ECDE outperformed traditional methods (e.g., NSGA-II) by over 20% in path efficiency and energy metrics, with robust obstacle avoidance and solution diversity. [Implementation and Impact] : Accepted by IEEE Transactions on Vehicular Technology (TVT). Integrated into smart city UAV modules for decision-making tasks. The algorithm is interpretable, scalable, and well-suited for embodied navigation across various urban scenarios.		
Huawei Technologies Co., Ltd. (Shanghai) – 2012 Lab Intern, Theoretical Research Department Shanghai		Jul. 2020 - Jan. 2021
Research on Intelligent Transportation Signal Restoration Algorithms: [Algorithm Innovation] : Developed a robust low-rank matrix completion model for high-missing-rate multi-source traffic data, combining ADMM with preconditioned conjugate gradient methods. Introduced a spatio-temporal kernel regularization term to capture dynamic road network structures, overcoming the convergence bottleneck of traditional non-convex optimization in large-scale scenarios. [Performance Enhancement] : Achieved over 95% imputation accuracy on real-world datasets including Guangzhou roads, Hangzhou metro, and Seattle highways — approximately 15% higher than baseline models — with a $10\times$ speed-up in convergence, supporting real-time repair of matrices with tens of millions of entries. [Implementation and Impact] : Core results were published in JCAM (JCR Q1) and integrated into Huawei’s communication network monitoring module, resulting in a 12.6% increase in peak-hour traffic efficiency, demonstrating strong practical deployment value.		

PH.D. RESEARCH HIGHLIGHTS

Ph.D. Dissertation: Research on Two Types of Hypercomplex Matrices Decomposition and Applications
Research Areas: Computational mathematics, matrix optimization, and high-performance algorithm design Core Contributions: [Algorithm Innovation] : Addressing the computational challenges in hypercomplex matrix structures (dual numbers and quaternions), this work proposes several efficient low-rank decomposition algorithms—such as DUTV, DQR, QUTV, and QCUR—with closed-form solutions and strong numerical stability, overcoming limitations of classical SVD in extended complex spaces. [Computational Optimization] : A unified framework combining randomized sampling and matrix blocking techniques was developed to significantly reduce both time and memory complexity while preserving decomposition accuracy. The algorithms are suitable for GPU-based parallel computing and edge deployment, with potential applications in accelerating preprocessing and inference tasks in deep learning systems—especially in large-scale scenarios such as visual intelligence and medical signal processing. [Implementation and Impact] : Five related papers have been published in top-tier journals including PJO, CAM, AML, NFAO, and LMA, covering both theoretical foundations and practical applications of hypercomplex matrix decompositions. The methods have been successfully applied to tasks such as fMRI-based brain imaging analysis and color video foreground separation, demonstrating strong generalization across domains.

SKILLS&INTERESTS

Computer Skills: Familiar with major programming languages such as Python, C, and MATLAB, as well as deep learning development workflows. Experienced in model training and deployment; knowledgeable about the PyTorch framework and GPU parallel computing. Capable of handling high-dimensional data and optimizing model performance, with prior experience in embedded AI application development.
Languages: Mandarin, English
Hobbies & Interests: Hiking (Certified Outdoor Sports Instructor, Chinese Mountaineering Association), Football (Level-3 Referee, Chinese Football Association), and Rowing.

Research Publications

Published Articles

- [1] **R Xu**, Z Huang, C Wang, and H Yan, "Evolving Collaborative Differential Evolution for Dynamic Multi-objective UAV Path Planning" in IEEE Transactions on Vehicular Technology, 2025 [**TOP Journal, Q1, IF: 7.1**]
- [2] **R Xu**, W Ma, M Che, "Numerical Radius Approach to Asymptotic Stability of Quaternion-Tensor Delay Systems" in Applied Mathematics Letters, 109795, 2025 [**TOP Journal, Q1, IF: 2.8**]
- [3] C. Wu, J. Cao, **R. Xu**, Z. Ran, M. Che, W. Zhu, and H. Yan, "DuSA: Fast and accurate dual-stage sparse attention mechanism accelerating both training and inference," in The Thirty-ninth Annual Conference on Neural Information Processing Systems, 2025. [**co-first author CCF A, TOP Conference**]
- [4] C. Wu, M. Che, **R. Xu**, Z. Ran, and H. Yan, "ELFATT: Efficient linear fast attention for vision transformers," in Proceedings of the 33rd ACM International Conference on Multimedia, ser. MM '25. New York, NY, USA: Association for Computing Machinery, 2025. [**co-first author CCF A, TOP Conference**]
- [5] **R Xu**, Y Wei, and H Yan, "LU Decomposition and Generalized Autoone-Takagi Decomposition of Dual Matrices and their Applications" in Linear and Multilinear Algebra 73.14: 3179-3201, 2025. [**Q2, IF: 1.2**]
- [6] **R Xu**, T Wei, Y Wei, and P Xie, "QR decomposition of dual matrices and its application" in Applied Mathematics Letters, 156, 109144, 2024. [**TOP Journal, Q1, IF: 2.8**]
- [7] **R Xu**, S Feng, Y Wei, and H Yan, "CUR and generalized CUR decompositions of quaternion matrices and their applications" in Numerical Functional Analysis and Optimization 45 (3), 234-258, 2024. [**Q2, IF: 1.4**]
- [8] **R Xu**, T Wei, Y Wei, and H Yan "UTV decomposition of dual matrices and its applications" in Computational and Applied Mathematics 43 (1), 41, 2024. [**TOP Journal, Q1, IF: 2.5**]
- [9] **R Xu**, Y Wei, "Randomized quaternion matrix UTV decomposition and its applications in quaternion matrix optimization" in Pacific Journal of Optimization, 20(2), 185-211, 2024. [**Q3, IF: 0.7**]
- [10] S Feng, **R Xu**, and Y Wei, "An ADMM-based Interior Point Method for Solving Nonnegative Tensor Least Squares Problems and its Applications" in Journal of Computational and Applied Mathematics, 472, 116793, 2025. [**TOP Journal, Q1, IF: 2.6**]
- [11] C Wang, **R Xu**, Y Huang, J Pei, C Huang, W Zhu, and J Yang, "Limited-Data SAR ATR Causal Method via Dual-Invariance Intervention" in IEEE Transactions on Geoscience and Remote Sensing, 2025. [**TOP Journal, Q1, IF: 8.6**]
- [12] S Feng, Y Wei, **R Xu**, and EK Chu, "A Linearized Alternating Direction Method of Multipliers for Solving Complex Semi-Symmetric Tensor Equations and its Applications" in Indian Journal of Pure and Applied Mathematics, 1-30, 2025. [**Q3, IF: 0.5**]
- [13] Z Xiong, Y Wei, **R Xu**, and Y Xu, "Low-rank traffic matrix completion with marginal information" in Journal of Computational and Applied Mathematics 410, 114219, 2022. [**TOP Journal, Q1, IF: 2.6**]
- [14] Z Ran, C Wu, **R Xu**, M Che, R Cheung, and H Yan, "FLAA: Fused Linear Attention Accelerator for Efficient Inference and Training in Transformers." In 2025 International Conference on Field Programmable Technology (ICFPT), Shanghai, China, IEEE, 2025. [**CCF C**]
- [15] Z Ran, C Wu, **R Xu**, M Che, and H Yan, "UniFormer: Unified and Efficient Transformer for Reasoning Across General and Custom Computing" in The Thirty-ninth Annual Conference on Neural Information Processing Systems Workshop on Efficient Reasoning, 2025. Available: <https://openreview.net/forum?id=RshUF8ArcQ>

Research Publications

In Preparation

- [1] **R Xu**, C Wu, Z Huang, W Zhu, Z Ran, and H Yan, "GP-STPCA: Generalized Power Method for Sparse Tensor Principal Component Analysis"
submitted to ICLR 2026
- [2] **R Xu**, C Wu, M Che, Z Ran, Y Wei, and H Yan, "sparseGeoHOPCA: A Geometric Solution to Sparse Higher-Order PCA Without Covariance Estimation"
submitted to ICLR 2026
- [3] Q Zhao, **R Xu**, Z Huang, Z Wang, and H Yan, "A Comprehensive Survey on Road Damage Detection: Datasets, Methods, and Platforms"
submitted to IEEE Transactions on Intelligent Transportation Systems, Major Revision
- [4] C Wang, C Wu, **R Xu**, Z Huang, C Yin, D Mao, S Zhang, and H Yan, "An Interpretable Two-Stage Feature Decomposition Method for Deep Learning-based SAR ATR"
submitted to IEEE Transactions on Image Processing
- [5] Q Zhao, Z Huang, **R Xu**, and H Yan, "Urban Air Mobility: Technological Foundations, Real-World Implementation, and the Future Development"
submitted to IEEE Transactions on Intelligent Transportation Systems
- [6] Z Ran, C Wu, **R Xu**, M Che, Z Wu, R Cheung, and H Yan, "HoC-CoT: High-Order Co-Clustering for Chain-of-Thought Reasoning"
submitted to AAAI 2025